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# SOLAR CELLS AND DEVICES USING TRIVALENT METAL DOPING

## Abstract

Provided is a solar cell utilizing trivalent metal doping and a method for manufacturing an inverted-structure organic solar cell (OSC), the method including: preparing a  $\text{TiO}_2$  precursor solution using titanium (IV) butoxide ( $\text{Ti}(\text{OCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{O})_4$ ); dissolving and stirring the  $\text{TiO}_2$  precursor solution in a mixture of 2-methoxyethanol ( $\text{CH}_3\text{OCH}_2\text{CH}_2\text{OH}$ ) and acetylacetone ( $\text{CH}_3\text{COCH}_2\text{COCH}_3$ ); dissolving  $\text{Ga}(\text{NO}_3)_3 \cdot x\text{H}_2\text{O}$  or  $\text{In}(\text{NO}_3)_3 \cdot x\text{H}_2\text{O}$  in the  $\text{TiO}_2$  precursor solution to prepare a doped  $\text{TiO}_2$  precursor solution. According to the method, it is possible to manufacture an inverted-structure OSC with improved power conversion efficiency (PCE) compared to conventional inverted-structure OSCs.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0032205, filed on Mar. 6, 2024, the disclosure of which is incorporated herein by reference in its entirety.

### BACKGROUND

#### 1. Field of the Invention

[0002] The present invention relates to a solar cell using trivalent metal doping. More specifically, the invention pertains to an inverted-structure organic solar cell with improved power conversion efficiency (PCE) by considering the impact assessment of trivalent metal doping on the performance of titanium dioxide ( $\text{TiO}_2$ ) as an electron transport layer.

#### 2. Discussion of Related Art

[0003] Organic solar cells (OSCs) have received substantial interest in recent years from both the academic and electronic industry sectors because to their high potential for use as low cost, flexible, lightweight, transparent, low dimension, and efficient light energy harvesters. Recently, the maximum power conversion efficiency (PCE) value of an OSC reached more than 18% as a result of tremendous efforts made over the past several years in areas like the fabrication of different semiconductors for using as donor acceptor, the synthesis of different semiconducting interlayers (hole extraction layer (HEL), electron transport layer (ETL), implementation of bandgap engineering, and device structure optimization. The basic OSC structure can be classified into two different types, namely conventional and inverted structures. There are several challenges associated with the conventional structured OSCs, including low environmental stability due to the etching effect of interlayer, easy oxidation of top electrode, and low photo stability. Inverted structures of OSCs have recently been devised to solve the issues associated with conventional structured OSCs due to greater device stability and manufacturing compatibility. Alongside these advantages, the inverted structured OSCs have a serious issue of a low PCE value, which is primarily due to poor electron transport ability of the n-type semiconductors that are utilized as the ETL in this device. Several inorganic oxide materials, including as tin oxide ( $\text{SnO}_2$ ), zinc oxide ( $\text{ZnO}$ ), and titanium oxide ( $\text{TiO}_2$ ) have been used as ETLs for inverted OSCs to increase electron collection at the indium tin oxide (ITO) cathode. Among these,  $\text{TiO}_2$  is the most commonly adopted ETL for photo- and air-stable OSCs owing to its chemical stability and non-toxicity; however, it exhibits a poor PCE value due to the considerably low conductivity of undoped  $\text{TiO}_2$ , which is a crucial issue that needs to be addressed. As a result, multiple research groups have investigated the utilization of different metal doped ETLs to increase the PCE of inverted OSCs. Metal doping is a viable strategy for changing grain size, conductivity, and orientation, and it has the ability to greatly affect the optoelectronic properties of ETLs. In this regard, the selection of the doping metal is very important because the atomic size varies for different metals; during doping, the lattice mismatch arises because of the difference in atomic size of Ti, and the doping metal may affect the charge transport ability of the resulting doped material. Previously, it was observed that Sn has a high potential for usage as a doping metal for  $\text{TiO}_2$  due to the minimal lattice misfit between  $\text{SnO}_2$  and  $\text{TiO}_2$ , which results in strong structural compatibility and stability. In 2014, Thambidurai reported that the PCE of a polymer donor based OSC can be improved (6.70% for  $\text{TiO}_2$  and 7.59% for Sn-doped  $\text{TiO}_2$ ) by doping its ETL  $\text{TiO}_2$  with Sn; this improvement can be attributed to the low shunt loss and

interfacial charge recombination provided by the  $\text{Sn}$ -doped  $\text{TiO}_2$  ETL based OSC. Recently, Kim et al. reported that the trivalent metal Al doped  $\text{TiO}_2$  ETL based OSC can exhibit an enhanced photoactive performance. Owing to the reduced interfacial recombination between the ETL and active layers (due to a highly crystalline upper photoactive layer), reduced trap density in  $\text{TiO}_2$  (through the elimination oxygen vacancy by Al), and widening of band gap (by Al doping), the Al doped  $\text{TiO}_2$  ETL based OSC exhibits a better PCE value than the undoped  $\text{TiO}_2$  based OSC. Other trivalent metals, such as Gallium (Ga) and Indium (In), have recently been used as doping reagents of various metal oxides for ETL applications, but to the best of our knowledge, very few trivalent metals have been evaluated as doping reagents of  $\text{TiO}_2$  for ETL applications. Furthermore, the effect of doping metal atomic size on  $\text{TiO}_2$  performance has not been investigated. Ga and In are two trivalent metals having different atomic size. Ga atom is significantly smaller than the In and Ti atom.

[0004] Therefore, in this study, first we studied the effect of doping by very little amount of Ga and In on the performance of  $\text{TiO}_2$  as ETL of Poly [[4,8-bis[(2-ethylhexyl)oxy]benzo[1,2-b:4,5-b']dithiophene-2,6-diyl][3-fluoro-2-[(2-ethylhexyl)carbonyl]thieno[3,4-b]thiophenediyl]]: Phenyl C71 butyric acid methyl ester (PTB7:PC:sub.70BM) active layer based inverted structured OSC. Following that, we investigated whether the atomic size of doping metals affects the performance of  $\text{TiO}_2$  as an ETL by comparing the performance of Ga and In doped  $\text{TiO}_2$  based OSCs. The test results showed that ETL doping (by trivalent metal) might considerably enhance the PCE of the inverted OSC. Furthermore, despite the fact that the Ga (187 pm) atom is substantially smaller than the In (220 pm) and Ti (215 pm) atoms, the PCE value of the Ga doped  $\text{TiO}_2$  ETL based inverted OSC was equivalent to the In doped  $\text{TiO}_2$  ETL based one.

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#### SUMMARY OF THE INVENTION

[0042] The technical problem of the present invention is to provide a method for manufacturing an inverted-structure organic solar cell (OSC) with improved power conversion efficiency (PCE) using trivalent metal doping.

[0043] To solve the aforementioned problem, the present invention provides an inverted-structure organic solar cell characterized by comprising an electron transport layer (ETL) made of titanium dioxide (TiO.sub.2) doped with a trivalent metal.

[0044] Additionally, the present invention provides an inverted-structure organic solar cell characterized in that the trivalent metal is indium (In). Furthermore, the present invention provides an inverted-structure organic solar cell characterized in that the trivalent metal is gallium (Ga). Moreover, the present invention provides an inverted-structure organic solar cell characterized by using PTB7:PC.sub.70BM as the active layer.

[0045] Additionally, the present invention provides a method for manufacturing an inverted-structure organic solar cell, comprising: A step of preparing a TiO.sub.2 precursor solution using titanium (IV) butoxide (Ti(OCH.sub.2CH.sub.2CH.sub.2CH.sub.3).sub.4); A step of dissolving and stirring the TiO.sub.2 precursor solution in a mixture of 2-methoxyethanol (CH.sub.3OCH.sub.2CH.sub.2OH) and acetylacetone (CH.sub.3COCH.sub.2COCH.sub.3); and A step of dissolving Ga(NO.sub.3).sub.3.Math.xH.sub.2O or In(NO.sub.3).sub.3.Math.xH.sub.2O in the TiO.sub.2 precursor solution to prepare a doped TiO.sub.2 precursor solution.

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## Description

#### BRIEF DESCRIPTION OF THE DRAWINGS

[0046] The above and other objects, features and advantages of the present invention will become more apparent to those of ordinary skill in the art by describing exemplary embodiments thereof in detail with reference to the accompanying drawings, in which:

[0047] FIG. 1 is a graph showing the XRD profiles of various films formed with TiO.sub.2, In-doped TiO.sub.2, and Ga-doped TiO.sub.2.

[0048] FIG. 2 shows SEM images and AFM images of various films formed with TiO.sub.2, In-doped TiO.sub.2, and Ga-doped TiO.sub.2.

[0049] FIG. 3 illustrates the device structure of the inverted-structure organic solar cell and the chemical structures of the donor and acceptor materials used in the active layer of the device.

[0050] FIG. 4 presents the J-V characteristic curves of various ETL-based OSCs operating under one-sun conditions and the EQE spectra of various ETL-based OSCs.

[0051] FIG. 5 is a graph showing the recombination characteristics and activation energy of OSCs based on different ETLs (undoped TiO.sub.2, In-doped TiO.sub.2, and Ga-doped TiO.sub.2).

#### DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0052] The objects and effects of the present invention will become clear through the following detailed description, but are not limited to the following description. In addition, in the description of the present invention, when it is determined that the detailed description of known technologies related to the present invention may unnecessarily obscure the gist of the present invention, the detailed description will be omitted.

[0053] Hereinafter, embodiments of the present invention will be described in detail with reference to the attached drawings so that those skilled in the art can easily implement the present invention. However, the present invention may be implemented in various forms and thus is not limited to the embodiments disclosed below. In addition, in order to clearly disclose the present invention in the drawings, parts not related to the present invention are omitted, and identical or similar symbols in the drawings indicate identical or similar components.

[0054] Inverted organic solar cells (OSCs) are becoming more popular because to their better durability and photo stability; nonetheless, their power conversion efficiency (PCE) is lower than that of conventional OSCs due to the poor conductivity of the electron transport layer (ETL). Several doped metal oxides have been investigated as possible ETLs to overcome this problem. Various metal doped titanium dioxide ( $\text{TiO}_2$ ) has recently been tested as an ETL for photo- and air-stable OSCs. Due to their physicochemical features, trivalent metals are one of the best possibilities as a  $\text{TiO}_2$  doping reagent. Therefore, we investigate at how  $\text{TiO}_2$ 's performance as an ETL of an inverted OSC may be affected by doping it with a very little quantity of trivalent metal in this study. Additionally, two other metals, such as Indium (In) and Gallium (Ga), are taken into consideration to test the impact of the doping metal's atomic size on  $\text{TiO}_2$ 's performance as an ETL. The doped- $\text{TiO}_2$  based OSC shows 10.24% higher PCE than undoped- $\text{TiO}_2$  based OSC. Furthermore, despite the fact that Ga's atomic size is substantially smaller than Ti's and In's, both Ga doped- $\text{TiO}_2$  based OSC (7.54%) and In-doped- $\text{TiO}_2$  (7.47%) display almost identical PCE values.

[0055] Before examining the inverted-structure organic solar cell utilizing trivalent metal doping according to an embodiment of the present invention with reference to the drawings, the manufacturing method will be described in detail.

#### Materials

[0056] Titanium (IV) butoxide ( $\text{Ti}(\text{OCH}_2\text{CH}_2\text{CH}_2\text{CH}_3)_4$ ), chlorobenzene ( $\text{C}_6\text{H}_5\text{Cl}$ ), 1,8-diiodooctane ( $\text{C}_8\text{H}_{16}\text{I}_2$ ), acetyl acetone ( $\text{CH}_3\text{COCH}_2\text{COCH}_3$ ), 2-methoxyethanol ( $\text{CH}_3\text{OCH}_2\text{CH}_2\text{OH}$ ), gallium (III) nitrate hydrate ( $\text{Ga}(\text{NO}_3)_3 \cdot x\text{H}_2\text{O}$ ), molybdenum trioxide ( $\text{MoO}_3$ ), indium (III) nitrate hydrate ( $\text{In}(\text{NO}_3)_3 \cdot x\text{H}_2\text{O}$ ), and anhydrous N,N-dimethylformamide (DMF, 99.8%) were supplied by Sigma-Aldrich. 1-Material, Inc. Ltd. supplied PTB7 and PC<sub>70</sub>BM. Indium tin oxide (ITO) coated substrates were collected from All for LAB. All of the chemicals were used without additional purification.

#### Solution Preparation

[0057] The PTB7:PC<sub>70</sub>BM (1:1.5) solution was prepared in a hybrid solvent of chlorobenzene/1, 8-diiodooctane (97:3 vol %) and stirred at room temperature for whole night in a nitrogen filled glove box.

#### Formation of ETL

[0058] In this work, the precursor solution for creating  $\text{TiO}_2$ -based ETL was made using 0.45 M of  $\text{Ti}(\text{OCH}_2\text{CH}_2\text{CH}_2\text{CH}_3)_4$ , which was mixed in a combination of  $\text{CH}_3\text{OCH}_2\text{CH}_2\text{OH}$  and 0.45 M of  $\text{CH}_3\text{COCH}_2\text{COCH}_3$ . A magnetic stirrer was used to mix the solution for 2 h.

[0059] Precursor solutions for Ga- and In-doped  $\text{TiO}_2$ -based ETL were produced by dissolving  $\text{Ga}(\text{NO}_3)_3 \cdot x\text{H}_2\text{O}$  and  $\text{In}(\text{NO}_3)_3 \cdot x\text{H}_2\text{O}$  in  $\text{TiO}_2$  precursor solutions, respectively. During the experiment, the doping ratio was set to 1 at %.

[0060] After that, thin layers of these mixtures were coated onto the pre-cleaned substrates (cleaned using acetone, isopropyl alcohol (IPA), and deionized (DI) water for 10 min each in an ultrasonic bath). The ETLs were formed through calcination at 450° C. for 2 h.

#### Measurement Equipment and Conditions

[0061] X-ray diffractometer (New D8 Advance) was used to record the XRD patterns of several

material-based ETL. The AFM, FESEM, and HRTEM images of different ETLs were captured by XE-100 AFM, Zeiss Supra 55VP SEM, and JEM-3010 TEM respectively. The In and Ga impurity concentration in the TiO<sub>2</sub> films was estimated by a Bruker Energy Dispersive X-ray Spectroscopy (EDX) machine. Keithley 237 source measurement (Newport 91160A) device and a 300 W solar simulator were used to test the J-V characteristics. Lock-in amplifier (Model: 7265, Signal Recovery) was used to measure the wavelength-dependent IPCE of OSCs. We used a SpectroPro-150 (Acton Research Corporation) Xenon lamp with a monochromator for illuminating (monochromatic light). In our study, the OSCs were tested in a vacuum chamber, and the temperature in the chamber was changed using a temperature controller (Lake Shore Cryotronics 331). The measurement started at 100 K and progressed in increments of 25 K until a final value of 350 K was reached. After fixing the temperature, the measurement lasted for 1 h to maintain it.

[0062] Hereinafter, the characteristics of the inverted-structure organic solar cell utilizing trivalent metal doping according to an embodiment of the present invention will be examined with reference to the drawings.

[0063] FIG. 1 is a graph illustrating the X-ray diffraction (XRD) profiles of various films formed with TiO<sub>2</sub>, In-doped TiO<sub>2</sub>, and Ga-doped TiO<sub>2</sub>.

[0064] After the deposition of different ETLs, we confirmed the formation of TiO<sub>2</sub>, and doped TiO<sub>2</sub> with the help of the observation of their X-ray diffraction (XRD) profiles (FIG. 1). The XRD profile of TiO<sub>2</sub> shows a wide diffraction peak at 25.36°, which reflects the (101) planes of TiO<sub>2</sub>'s tetragonal wurtzite structure, as seen in FIG. 1. For the In and Ga doped TiO<sub>2</sub>, the angle associated with the peak (101) has shifted to 25.37° and 25.50°, respectively. Moreover, from the XRD profiles, we estimated the average grain size (D) of both undoped and doped TiO<sub>2</sub> crystals by employing the Scherrer equation (Eq. (1)),

$$[00001] D = \frac{K \times \lambda}{\cos \beta} \quad (\text{Eq. 1})$$

[0065] where K, λ, and β is Scherrer constant, wavelength of X-ray, and full width of half maximum of the diffraction angle peak respectively. The value of λ is 1.54 Å for Cu Kα1.

[0066] On the other hand, Bragg's law was employed for estimating the lattice spacing, d (Eq. 2),

$$[00002] n \lambda = 2d \sin \theta \quad (\text{Eq. 2})$$

[0067] where n is integer (+).

[0068] From the analysis of XRD profile of different films, the average grain size and 'd' of undoped TiO<sub>2</sub> were estimated to be 12.72 nm and 3.42 Å respectively (JCPDS card no. 89-4921); whereas, for In doped TiO<sub>2</sub> film, they were found to be 12.68 nm and 3.28 Å, respectively. The proximity between the D and d values of In doped TiO<sub>2</sub> and undoped TiO<sub>2</sub> films is due to the similar atomic size of In (215 pm) and Ti (220 pm).

[0069] On the other hand, the Ga doped TiO<sub>2</sub> exhibited a significant decrement in both the average grain size (11.64 nm) and 'd' value (2.14 Å). The atomic size of Ga is 187 pm, which is significantly lower than that of Ti, and owing to this size mismatch, the estimated average grain size and d value of Ga doped TiO<sub>2</sub> are slightly lower than those of undoped TiO<sub>2</sub>.

[0070] FIG. 2 presents SEM (Scanning Electron Microscopy) and AFM (Atomic Force Microscopy) images of various films formed with TiO<sub>2</sub>, In doped TiO<sub>2</sub>, and Ga-doped TiO<sub>2</sub>.

[0071] Scanning electron microscopy (SEM) images of TiO<sub>2</sub>, In doped TiO<sub>2</sub>, and Ga doped TiO<sub>2</sub> ETLs are shown in (a), (b) and (c) of FIG. 2 respectively. The SEM images of various films show that the grains are spread consistently throughout the surface of the substrate, and all of the films are uniform and crack-free. The findings are supported by AFM images ((d), (e), and (f) of FIG. 2) of those films.

[0072] Furthermore, root-mean-square of the surface roughness of an undoped TiO<sub>2</sub> based ETL is 2.91 nm, but it is 2.92 nm and 2.59 nm for In- and Ga doped TiO<sub>2</sub> films, respectively, based on AFM images of the films developed by the various samples. Owing to the proximate

average grain sizes of undoped TiO<sub>2</sub> and In doped TiO<sub>2</sub>, the film formed by these materials exhibit similar surface roughness. However, the Ga doped TiO<sub>2</sub> has a smaller grain size than the undoped TiO<sub>2</sub>, and therefore, its surface roughness value is lower than that of the undoped TiO<sub>2</sub> film. The EDX spectra of In and Ga doped TiO<sub>2</sub> films confirm the presence of the doping elements (In or Ga) within the TiO<sub>2</sub> layer.

[0073] Furthermore, elemental analysis shows that 0.76 at. % Ga is present in the Ga doped TiO<sub>2</sub> film, whereas 0.62 at. % In is present in the In doped TiO<sub>2</sub> film. To compare the potentiality of different ETLs for the case of inverted structured OSC, we fabricated different inverted structured OSCs having PTB7:PC<sub>70</sub>BM active layer with TiO<sub>2</sub>, In doped TiO<sub>2</sub>, and Ga doped TiO<sub>2</sub> as ETL.

[0074] FIG. 3 is a schematic diagram illustrating the device structure of the inverted-structure organic solar cell (OSC) and the chemical structures of the donor and acceptor materials used in the active layer. FIG. 4 presents the J-V characteristic curves of various ETL-based OSCs operating under one-sun conditions and the EQE spectra of various ETL-based OSCs.

[0075] Specifically, referring to (a) of FIG. 3, a schematic diagram of an inverse-structure OSC in which doped TiO<sub>2</sub> is used as an electron transport layer (ETL), a PTB7:PC<sub>70</sub>BM active layer is formed, and molybdenum oxide (MoO<sub>3</sub>) is used as a hole extraction layer (HEL) can be confirmed. In addition, referring to (b) of FIG. 3, the chemical structures of PTB7 as a donor of the active layer (PTB7:PC<sub>70</sub>BM) and PC<sub>70</sub>BM as an acceptor can be confirmed, respectively.

[0076] (a) of FIG. 4 depicts the current density (J)-(V) characteristics of OSCs with ETL formed by TiO<sub>2</sub>, In doped TiO<sub>2</sub>, and Ga doped TiO<sub>2</sub>. The photovoltaic performance parameters (along with standard deviated error) are summarized in Table 1.

TABLE-US-00001 TABLE 1 Summary of device performance parameters (averaged for eight devices) of different ETL-based inverted structured OSCs for 1-sun condition.

| ETL Parameter                          | TiO <sub>2</sub> | In TiO <sub>2</sub> | Ga TiO <sub>2</sub> |
|----------------------------------------|------------------|---------------------|---------------------|
| J <sub>SC</sub> (mA cm <sup>-2</sup> ) | 14.80 ± 0.04     | 15.20 ± 0.02        | 15.52 ± 0.03        |
| V <sub>OC</sub> (V)                    | 0.75 ± 0.01      | 0.76 ± 0.02         | 0.77 ± 0.06         |
| FF (%)                                 | 61.86 ± 2.1      | 64.56 ± 1.10        | 64.78 ± 0.32        |
| PCE (%)                                | 6.84 ± 0.09      | 7.47 ± 0.16         | 7.54 ± 0.14         |
| R <sub>s</sub> (Ωcm <sup>2</sup> )     | 7.61 ± 0.51      | 6.15 ± 0.07         | 6.20 ± 0.11         |
| R <sub>sh</sub> (Ωcm <sup>2</sup> )    | 1001.65 ± 5.01   | 1317.80 ± 8.01      | 1302.28 ± 10.01     |

[0077] The values of average open circuit voltage (VOC), short circuit current density (JSC), fill factor (FF), and PCE for undoped TiO<sub>2</sub> ETL based OSC is 0.75 V, 14.80 mAcm<sup>-2</sup>, 61.86%, and 6.84%, respectively; the values for In doped TiO<sub>2</sub> ETL based OSC are 0.76 V, 15.20 mAcm<sup>-2</sup>, 64.50%, and 7.46%, respectively; the values for Ga doped TiO<sub>2</sub> ETL based OSC are 0.77 V, 15.52 mAcm<sup>-2</sup>, 64.78%, 7.54%, respectively. Notably, the performance of the inverted structured OSC can be improved significantly by doping its ETL with both In and Ga. Additionally, the performance of Ga doped TiO<sub>2</sub> ETL based one is substantially identical in all aspects to that of In doped TiO<sub>2</sub> ETL based one.

[0078] The incident photon to charge carrier efficiency (IPCE) spectra ((b) of FIG. 4) of different ETL based OSCs also exhibits same variation trend. The recombination coefficient (α) of OSC is an important parameter which depends on the light power (PLight) (Eq. (3)),

$$J_{SC} = P_{Light} \quad (\text{Eq. 3})$$

[0079] where α is the recombination coefficient.

[0080] FIG. 5 is a graph illustrating the recombination characteristics and activation energy of organic solar cells (OSC) based on different electron transport layers (ETLs), including undoped TiO<sub>2</sub>, In-doped TiO<sub>2</sub>, and Ga-doped TiO<sub>2</sub>.

[0081] For the estimation of α, we recorded the J-V characteristics curve of the OSCs at various temperatures in the range 100-350 K. (a) of FIG. 5 illustrates the variation of α of different ETL based OSCs with temperature. From (a) of FIG. 5, it is observed that the α of undoped TiO<sub>2</sub> ETL based OSC is approximately 0.8-0.9 within the temperature range, and because of doping of

ETL with In or Ga, the value of  $\Delta$  of the OSC has increased noticeably (approximately 0.92-1.00). Furthermore, there is no significant variation in the  $\Delta$  value when different metals (In, Ga) are used for doping. The doping of the ETL results in an increase in the recombination coefficient of OSC, which indicates an improvement in charge extraction (with less space charge effect) within the device. This outcome is also attributed to the In or Ga-doped TiO<sub>2</sub> layer's improved interfacial energy level alignment, which helps to balance electron and hole transport.

[0082] Another essential parameter that describes the potential barrier height associated with the charge transport across interlayers is activation energy ( $A$ ). Smaller  $A$  is more favorable for achieving better device performance. It is composed of two parts: the band gap (energy required to excite electrons from HOMO to LUMO) and the energy required to delocalize the charge carriers (i.e. the energy required for an electron to escape from a trap). Phonons can also aid in the delocalization process in organic materials. At low temperatures, the likelihood of finding a phonon with enough energy to permit a hop to the adjacent site is minimal. Charge carrier mobility, and so conductivity, should increase as temperature rises. As a general conclusion for organic materials, at low temperatures, few charge carriers are thermally produced and are not easily transferred; hence, the current is minimal in the absence of light.

[0083] The value of  $\Delta$  can be calculated using temperature dependent JSC data at different light intensity. The J-V curves of TiO<sub>2</sub>, In doped TiO<sub>2</sub>, and Ga doped TiO<sub>2</sub> ETLs based OSCs at temperatures ranging from 100 to 350 K and light intensities ranging from 10 to 100 mW/cm<sup>2</sup>, shows that the device performance of doped TiO<sub>2</sub> ETL-based OSC is superior to that of undoped TiO<sub>2</sub> based devices. The JSC of OSCs grew linearly as temperature increased, whereas VOC rose as temperature decreased since VOC is inversely related to temperature. The  $\Delta$  value is estimated using Eq. (4) and the results of the J-V characteristics at different temperature and light intensities:

$$[00004] J_{SC}(T, P_{\text{Light}}) = J_0(P_{\text{Light}}) \exp\left(\frac{P_{\text{Light}}}{K_B \times T}\right) \quad (\text{Eq. 4})$$

[0084] (b) of FIG. 5 illustrates the variation of  $\Delta$  of OSCs having different ETLs with light intensity. In (b) of FIG. 5, it is interesting to observe that the  $\Delta$  value in doped ETL (In or Ga doped TiO<sub>2</sub>) based OSCs is lower than that of undoped ETL (TiO<sub>2</sub>) based OSC. This suggests that the OSC's ETL can be doped with a trivalent metal to minimize the potential barrier between its interlayers, and as a result, the OSC with the doped TiO<sub>2</sub> based ETL can perform better than the undoped TiO<sub>2</sub> based one. Furthermore, the effect of atomic size of the doping metal on the decrement in the potential barrier is negligible. Because of this, it's likely that the Ga doped TiO<sub>2</sub> based OSC performed similarly to the In doped TiO<sub>2</sub> ETL based OSC (Table 1) although having much lower atomic sizes (Ga (187 pm) compared to Ti (215 pm) and In (220 pm), respectively). All of the foregoing data indicate that doping TiO<sub>2</sub> with trivalent metal can greatly increase its performance as an ETL, but there is no significant effect of doping metal on its performance as an ETL.

[0085] In conclusion, we investigated the influence of small amounts of trivalent metal doping on the potential of TiO<sub>2</sub> as the ETL of a PTB7:PC<sub>71</sub>BM active layer-based inverted structured OSC. It was observed that the undoped TiO<sub>2</sub> based OSC showed 6.84% PCE; whereas the Ga doped TiO<sub>2</sub> based OSC showed 7.54% PCE, which similar to the PCE value exhibited by the In doped TiO<sub>2</sub> based OSC (7.47%). Doping with trivalent metals In or Ga lowered the density of electron trap state to the TiO<sub>2</sub> lattice by eliminating oxygen vacancies. This improved TiO<sub>2</sub>'s electron extractability, and as a result, the doped TiO<sub>2</sub> ETL-based OSCs had a much higher PCE value than the undoped TiO<sub>2</sub>-based OSCs. However, despite the fact that the atomic size of Ga is significantly smaller than that of In and Ti, no significant effect of doping metal atomic size on the performance of TiO<sub>2</sub> as ETL was observed because this phenomenon has no effect on the charge transport characteristic (recombination coefficient, activation energy, etc.) of the entire system.

[0086] According to the present invention, it is possible to manufacture an inverted-structure OSC with improved PCE compared to conventional inverted-structure OSCs.

[0087] The preferred embodiments of the present invention described above are disclosed for illustrative purposes, and those skilled in the art will be able to make various modifications, changes and additions within the spirit and the scope of the present invention, which should be considered as falling within the scope of the patent claims. In addition, since those skilled in the art to which the present invention pertains can make various substitutions, modifications and changes without departing from the technical spirit of the present invention, the present invention is not limited to the above-described embodiments and attached drawings.

[0088] In the above-described exemplary system, the methods are described on the basis of a flowchart as a series of steps or blocks, but the present invention is not limited to the order of the steps, and some steps may occur in a different order or may occur simultaneously as described above. In addition, those skilled in the art will understand that the steps described in the flowchart are not exclusive and that other steps may be included or one or more steps in the flowchart may be deleted without affecting the scope of the present invention.

## Claims

1. An inverted-structure organic solar cell (OSC) comprising an electron transport layer (ETL) made of trivalent metal-doped titanium dioxide (TiO<sub>2</sub>).
  2. The inverted-structure organic solar cell of claim 1, wherein the trivalent metal is indium (In).
  3. The inverted-structure organic solar cell of claim 1, wherein the trivalent metal is gallium (Ga).
  4. The inverted-structure organic solar cell of claim 1, wherein the active layer comprises PTB7:PC<sub>71</sub>BM.
  5. The inverted-structure organic solar cell of claim 1, wherein the hole extraction layer (HEL) comprises molybdenum trioxide (MoO<sub>3</sub>).
  6. A method for manufacturing an inverted-structure organic solar cell, comprising: preparing a TiO<sub>2</sub> precursor solution using titanium (IV) butoxide (Ti(OCH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>3</sub>)); dissolving and stirring the TiO<sub>2</sub> precursor solution in a mixture of 2-methoxyethanol (CH<sub>3</sub>OCH<sub>2</sub>CH<sub>2</sub>OH) and acetylacetone (CH<sub>3</sub>COCH<sub>2</sub>COCH<sub>3</sub>); and dissolving Ga(NO<sub>3</sub>)<sub>3</sub>·xH<sub>2</sub>O or In(NO<sub>3</sub>)<sub>3</sub>·xH<sub>2</sub>O in the TiO<sub>2</sub> precursor solution to prepare a doped TiO<sub>2</sub> precursor solution.
  7. The method for manufacturing an inverted-structure organic solar cell of claim 6, wherein the doping ratio of Ga(NO<sub>3</sub>)<sub>3</sub>·xH<sub>2</sub>O or In(NO<sub>3</sub>)<sub>3</sub>·xH<sub>2</sub>O is 1 at. %.
  8. The method for manufacturing an inverted-structure organic solar cell of claim 6, wherein the TiO<sub>2</sub> precursor solution is dissolved and stirred in a mixture of 2-methoxyethanol (CH<sub>3</sub>OCH<sub>2</sub>CH<sub>2</sub>OH) and acetylacetone (CH<sub>3</sub>COCH<sub>2</sub>COCH<sub>3</sub>) for 2 hours using a magnetic stirrer; and wherein the TiO<sub>2</sub> precursor solution is calcined at 450° C. for 2 hours to fabricate an electron transport layer (ETL) made of titanium dioxide (TiO<sub>2</sub>).
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